

Tableau Notes: Granularity, Aggregation, Discrete & Continuous

1. Granularity in Tableau

What is Granularity?

Granularity refers to the **level of detail (LOD)** at which your data is displayed in a visualization.

Think of it as:

"How detailed is the data shown in the view?"

The dimensions you place on the Rows, Columns, or Marks card determine the granularity.

Example Dataset

ORDER ID	CATEGORY	CUSTOMER	ORDER DATE	SALES
101	Furniture	Alice	1 Jan	500
102	Furniture	Bob	1 Jan	700
103	Technology	Alice	2 Jan	1000
104	Furniture	Alice	3 Jan	300

Example 1: Category Granularity

Rows:

Category

Columns:

SUM(Sales)

Output:

CATEGORY	SALES
FURNITURE	1500
TECHNOLOGY	1000

The granularity is **Category** because Tableau groups all records by category.

Example 2: Category + Customer Granularity

Rows:

Category

Customer

Output:

CATEGORY	CUSTOMER	SALES
FURNITURE	Alice	800
FURNITURE	Bob	700
TECHNOLOGY	Alice	1000

Now the granularity becomes **Category + Customer**, giving a more detailed view.

Key Point

Adding more dimensions **increases the level of detail**.

Example progression:

Category

↓

Category + Customer

↓

Category + Customer + Order Date

Each additional dimension increases granularity.

2. Aggregation

What is Aggregation?

Aggregation means **combining multiple values into a single summarized value**.

Tableau automatically aggregates **Measures**.

Common aggregations include:

- SUM
 - AVG
 - MIN
 - MAX
 - COUNT
 - COUNTD
-

Example

Suppose the Sales values are:

Sales

100

200

300

Different aggregations produce:

SUM = 600

AVG = 200

MIN = 100

MAX = 300

COUNT = 3

Example in Tableau

Rows:

Category

Columns:

SUM(Sales)

Output:

Category	Sales
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Furniture	1500
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Technology	1000
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Instead of showing every sales value, Tableau **aggregates** them into a total.

3. Aggregate Measures (ON vs OFF)

By default, Tableau **aggregates measures**.

Aggregate Measures ON (Default)

If the data is:

ORDER ID	SALES
101	100
102	200
103	300

Dragging **Sales** into the view shows:

SUM(Sales)

600

Only one aggregated value appears.

Aggregate Measures OFF

You can turn aggregation off to see **each individual record**.

Steps:

1. Go to the **Analysis** menu.
2. Uncheck **Aggregate Measures**.

Analysis

✓ Aggregate Measures

After unchecking it:

ORDER ID	SALES
101	100
102	200
103	300

Now Tableau displays each row separately instead of summarizing them.

Why Turn Off Aggregate Measures?

Use it when you want to:

- View every record in the dataset.
- Check for duplicate rows.
- Validate imported data.
- Analyze data at the row level.
- Build scatter plots where each mark represents a single record.

Note: Turning off **Aggregate Measures** does not change the underlying data. It only changes how Tableau displays the measures in the current view.

4. Discrete vs Continuous

This is one of the most important Tableau concepts.

Discrete

A **Discrete** field contains **distinct, separate values**.

It creates **headers** (individual categories).

Displayed as **blue** in Tableau.

Examples:

- Category
- Region
- Customer Name

- Order ID
- Month (Discrete)

Example

Category:

Furniture

Technology

Office Supplies

These are separate categories.

Visualization:

Furniture

Technology

Office Supplies

Each category gets its own header.

Continuous

A **Continuous** field represents a **range of values**.

It creates an **axis**.

Displayed as **green** in Tableau.

Examples:

- Sales
 - Profit
 - Discount
 - Date (Continuous)
-

Example

Sales values:

100

200

300

400

500

Instead of separate headers, Tableau creates an axis:

100 ---- 200 ---- 300 ---- 400 ---- 500

Date Field Example (Discrete vs Continuous)

Suppose you have the following orders:

ORDER DATE	SALES
1 JAN	100
5 JAN	200
12 JAN	300
20 JAN	250
28 JAN	400

Discrete Date

If you drag **Order Date** and select:

MONTH(Order Date)

(Discrete)

Tableau treats months as **categories**.

Output:

MONTH	SALES
JANUARY	1250

Or, across multiple months:

January

February

March

April

Each month is a separate header.

Continuous Date

If you choose:

MONTH(Order Date)

(Continuous)

Tableau creates a **continuous timeline**.

Jan ----- Feb ----- Mar ----- Apr

The axis represents time continuously, making it ideal for line charts and trend analysis.

Another Example

Discrete

Year

2022

2023

2024

Separate headers.

Continuous

2022 ----- 2023 ----- 2024

A continuous time axis.

When to Use Each

Use Discrete when:

- Creating tables.
- Comparing categories.
- Building bar charts.
- Grouping data into buckets.
- Showing individual years, months, or categories as separate headers.

Use Continuous when:

- Creating line charts.
- Showing trends over time.
- Working with numeric ranges.
- Displaying a continuous timeline for dates.

Blue vs Green in Tableau

PROPERTY	DISCRETE	CONTINUOUS
COLOR	Blue	Green
DISPLAY	Headers	Axis
VALUES	Separate categories	Continuous range
USED FOR	Bar charts, tables	Line charts, trend analysis
EXAMPLE	Category, Region, Month (Discrete)	Sales, Profit, Month (Continuous)

Summary

CONCEPT	MEANING	EXAMPLE
GRANULARITY	The level of detail shown in the view, determined by the dimensions used.	Category vs Category + Customer
AGGREGATION	Summarizing multiple values into one using functions like SUM, AVG, or COUNT.	SUM(Sales) by Category
AGGREGATE MEASURES (OFF)	Shows individual records instead of aggregated values.	Go to Analysis → uncheck Aggregate Measures to display row-level data.
DISCRETE	Separate, distinct values that create headers (blue).	Category, Region, Month (Discrete)

CONTINUOUS

Values on a continuous scale that create an axis (green).

Sales, Profit, **Month (Continuous)**